MAT 221 - CALCULUS I
Lesson 23 - 25 - Convergence and Divergence of Series

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Test 1: Divergence test

Theorem 1

Divergence Test

If $\lim_{n\to\infty} a_n \neq 0$ *then* $\sum a_n$ *will diverge.*

Example 1

Use divergence test to determine convergence or divergence of the following

$$(a)\ \sum_{n=1}^{\infty} 8\left(\frac{3}{4}\right)^n \qquad (b)\ \sum_{n=1}^{\infty} \frac{n}{n+2} \qquad (c)\ \sum_{n=1}^{\infty} \frac{1}{n} \qquad (d)\ \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

Solution

In class

Test 2: Ratio Test

Theorem 2

Ratio Test

Suppose we have the series $\sum a_n$, where a_n are positive terms. Define,

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$$

Then,

1. *if $L < 1$ the series is absolutely convergent (and hence convergent).*
2. *if $L > 1$ the series is divergent.*
3. *if $L = 1$ test fails.*

Example 2

Use ratio test to determine convergence or divergence of the following

$$(a)\ \sum_{n=1}^{\infty} \frac{2n-1}{2^n} \qquad (b)\ \sum_{n=1}^{\infty} \frac{3^{n-1}}{4n+1} \qquad (c)\ \sum_{n=1}^{\infty} \frac{2n^2+1}{3n^2+2} \qquad (d)\ \sum_{n=1}^{\infty} \frac{n!}{5^n}$$

Solution

In class

Test 3: Root Test

Theorem 3

Root Test

Suppose we have the series $\sum a_n$, where a_n are positive terms. Define,

$$L = \lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} (a_n)^{\frac{1}{n}}$$

Then,

1. *if $L < 1$ the series is absolutely convergent (and hence convergent).*
2. *if $L > 1$ the series is divergent.*
3. *if $L = 1$ test fails.*

Fact

$$\lim_{n\to\infty} n^{\frac{1}{n}} = 1$$

Example 3

Use root test to determine convergence or divergence of the following

$$(a)\ \sum_{n=1}^{\infty} \frac{n^2}{2^n} \qquad (b)\ \sum_{n=1}^{\infty} \left(\frac{5n-3n^3}{7n^3+2}\right)^n \qquad (c)\ \sum_{n=1}^{\infty} \frac{2^n}{n^3} \qquad (d)\ \sum_{n=1}^{\infty} \left(\frac{1}{1+n}\right)^n$$

Test 4: Integral Test

Theorem 4

Suppose that $f(x)$ is a continuous, positive and decreasing function on the interval $[k, \infty)$ and that $f(n) = a_n$ then,

1. *if* $\int_k^{\infty} f(x)\,dx$ *is convergent so is* $\sum_{n=k}^{\infty} a_n$
2. *if* $\int_k^{\infty} f(x)\,dx$ *is divergent so is* $\sum_{n=k}^{\infty} a_n$

Example 4

Use integral test to determine convergence or divergence of the following

$$(a)\ \sum_{n=1}^{\infty} \frac{1}{n} \qquad (b)\ \sum_{n=0}^{\infty} ne^{-n^2} \qquad (c)\ \sum_{n=1}^{\infty} \frac{n}{n^2+4} \qquad (d)\ \sum_{n=2}^{\infty} \frac{1}{n\ln(n)}$$

Solution

In class

Test 5: The p-Series Test

Theorem 5

If $k > 0$ then $\sum_{n=k}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leqslant 1$.

Example 5

Determine convergence or divergence of the following

$$(a) \sum_{n=1}^{\infty} \frac{1}{n^2} \qquad (b) \sum_{n=1}^{\infty} \frac{1}{n^7} \qquad (c) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \qquad (d) \sum_{n=1}^{\infty} n^{-\frac{2}{3}}$$

Solution

In class

Test 6: GP Series Test

Theorem 6

If $k > 0$, $a \in \mathbb{R}$ *and* $r \in \mathbb{R}$ *then* $\sum_{n=k}^{\infty} ar^n$ *converges if* $|r| < 1$ *and diverges if* $|r| \geqslant 1$.

Example 6

Determine convergence or divergence of the following

$$(a)\ \sum_{n=1}^{\infty} \frac{4}{3^n} \qquad (b)\ \sum_{n=1}^{\infty} \frac{1}{5^{n-1}} \qquad (c)\ \sum_{n=1}^{\infty} 2^{n+3} \qquad (d)\ \sum_{n=1}^{\infty} 5(3)^{2n-3}$$

Solution

In class

Test 7: Alternating Series Test

Theorem 7

Suppose that we have a series $\sum a_n$ and either $a_n = (-1)^n b_n$ or $a_n = (-1)^{n+1} b_n$ where $b \geqslant 0$ for all n. Then if,

1. $\lim_{n \to \infty} b_n = 0$ *and,*
2. $\{b_n\}$ *is a deceasing sequence the series $\sum a_n$ is convergent.*

Example 7

Use alternating series test to determine convergence or divergence of the following

$$(a) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \quad (b) \sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n^2+5} \quad (c) \sum_{n=1}^{\infty} \frac{(-1)^{n-3}\sqrt{n}}{n+4} \quad (d) \sum_{n=2}^{\infty} \frac{\cos(n\pi)}{\sqrt{n}}$$

Solution

In class

Test 8: Comparison Test

Theorem 8

Suppose that we have a series $\sum a_n$ and $\sum b_n$ with a_n, $b_n \geqslant 0$ for all n and $a_n \leqslant b_n$ for all n. Then,

1. *if $\sum b_n$ is convergent then so is $\sum a_n$.*
2. *If $\sum a_n$ is divergent then so is $\sum b_n$ is convergent.*

Example 8

Use comparison test to determine convergence or divergence of the following

$$(a)\ \sum_{n=1}^{\infty} \frac{1}{n^n} \quad (b)\ \sum_{n=1}^{\infty} \frac{n}{n^2 - \cos^2(n)} \quad (c)\ \sum_{n=1}^{\infty} \frac{1}{n(n+1)} \quad (d)\ \sum_{n=1}^{\infty} \frac{n^2+2}{n^4+5}$$

Solution

In class

Test 9: Limit Comparison Test

Theorem 9

Suppose that we have two series $\sum a_n$ *and* $\sum b_n$ *with* $a_n,\ b_n \geqslant 0$ *for all* n. *Define,*

$$c = \lim_{n\to\infty} \frac{a_n}{b_n}$$

If $c > 0$ *and is finite, then either both series converge or both series diverge.*

Example 9

Use limit comparison test to determine convergence or divergence of the following

$$(a)\ \sum_{n=0}^{\infty} \frac{1}{3^n - n} \quad (b)\ \sum_{n=1}^{\infty} \frac{4n^2 - 1}{n^3 + 9} \quad (c)\ \sum_{n=2}^{\infty} \frac{4n^2 + n}{\sqrt[3]{n^7 + n^3}} \quad (d)\ \sum_{n=7}^{\infty} \frac{4}{n^2 - 2n - 3}$$

Solution

In class

Test 10: Absolute convergence and Conditional convergence

In the previous Tests, we have been dealing with $\sum_{n=1}^{\infty} a_n$, where a_n are positive terms.

Definition 10

A series $\sum a_n$ is called **absolutely convergent** if $\sum |a_n|$ is convergent. If $\sum a_n$ is convergent and $\sum |a_n|$ is divergent we call the series **conditionally convergent**

Fact

If $\sum a_n$ is absolutely convergent then it is also convergent.

Example 10

Determine if each of the following series are absolute convergent, conditionally convergent or divergent

$$(a)\ \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \qquad (b)\ \sum_{n=1}^{\infty} \frac{(-1)^{n+2}}{n^2} \qquad (c)\ \sum_{n=1}^{\infty} \frac{\sin(n)}{n^3}$$

Solution

In class